MUHAMMAD HAMZA AKRAM

Senior Full-Stack Software Engineer | React, Next.js, Node.js, AI Automation

Karachi, Pakistan | (+92) 315-3182758 | hamzakram.dev@gmail.com
hamza-akram.vercel.app | linkedin.com/in/hamza-akram-dev | github.com/hkakram97

PROFESSIONAL SUMMARY

Senior Software Engineer with 6+ years of experience turning complex ideas into fast, scalable web, mobile, and AI products. Full-stack capability across frontend (React, Next.js, TypeScript, React Native), backend (Node.js, NestJS, Express.js, Django, REST APIs), and applied AI (LLM-powered workflows, voice agents, automation pipelines). Delivered 30+ production projects, including a hotel management platform shipped end-to-end in 6 weeks, an API optimization cutting response time from 2+ minutes to under 1 second, and an AI voice-agent automation platform for healthcare scheduling. Experienced leading engineering initiatives, owning features end-to-end from architecture through deployment, and working across cloud infrastructure (AWS) and databases (PostgreSQL, Redis, MongoDB).

CORE SKILLS

Frontend: React.js, Next.js, TypeScript, JavaScript (ES6+), React Native, Expo, Redux, RTK, Astro.js, Svelte, Angular, Tailwind CSS, Material UI, Bootstrap, Shadcn, Daisy UI, HTML, CSS

Backend: Node.js, NestJS, Express.js, Django, Django REST Framework, Python, PHP, REST API design, django-tenants, Celery, multi-tenant architecture

AI & Automation: LLM-powered workflows, AI voice agents, ElevenLabs, n8n automation, Pinecone, Cursor AI, Google APIs (Calendar, Sheets), prompt-driven workflow design

Cloud & DevOps: AWS, AWS CDK, ECS Fargate, Vercel, CI/CD, application performance and scalability optimization, caching, database indexing

Databases: PostgreSQL, MySQL, MongoDB, Firestore, Redis, Elasticsearch

CMS & E-commerce: WordPress, WooCommerce, Shopify, Magento, Joomla, Sanity.io headless CMS

Tools & Practices: Git, GitHub, Bitbucket, code reviews, agile collaboration, cross-functional team leadership

Languages: English, Urdu, Arabic, Chinese

PROFESSIONAL EXPERIENCE

Senior Software Engineer March 2023 – Present

Techwards, Karachi

- Develop AI-powered automation and voice-agent solutions integrating LLM-powered workflows, ElevenLabs, n8n, Google APIs, and scheduling systems.
- Lead development initiatives and collaborate with cross-functional teams to deliver production-ready features within demanding timelines.
- Optimize application performance, scalability, security, and reliability through database optimization, caching, pagination, infrastructure improvements, and architectural enhancements.
- Build modern web applications using React and Next.js and cross-platform mobile applications using React Native and Expo.
- Design and develop scalable web and mobile applications across SaaS, healthcare, e-commerce, education, publishing, and business-management domains.
- Own features end-to-end, including technical design, backend architecture, API development, frontend integration, database optimization, cloud deployment, and production improvements.
- Build backend services and REST APIs using Node.js, TypeScript, NestJS, Express.js, Django, PostgreSQL, Redis, and AWS.

Software Engineer March 2021 – February 2023

Techwards, Karachi

- Developed and maintained production web applications across multiple domains, including healthcare, e-commerce, education, SaaS, and customer-facing platforms.
- Built reusable frontend features using React, Angular, Next.js, and modern state-management libraries.
- Developed backend integrations and application features using Python, Django, PHP, and Node.js.
- Worked with PostgreSQL and other data-driven systems while contributing to API development, application optimization, and third-party integrations.
- Collaborated with larger engineering teams to deliver features, resolve production issues, and improve application usability and performance.

Web Developer **November 2018 – October 2020**

Innoventure, Karachi

- Developed and maintained WordPress websites for e-commerce and corporate businesses, focusing on responsive design and usability.
- Built e-commerce websites that supported online sales and customer acquisition.
- Developed a multi-country website for a pharmaceutical company, supporting localized content and business requirements.
- Worked with clients to translate business requirements into functional, user-friendly web experiences.

Web Developer (Early Career) **June 2017 – September 2017**

MATZ Solution, Karachi

- Built foundational experience in web development, responsive design, and website development workflows.
- Developed multiple small-scale websites while learning and applying modern web development practices.
- Built a survey management system supporting survey creation, data collection, and result management.

KEY PROJECTS

Horse Riding Management Platform — UAE

React.js, Express.js, PostgreSQL, Redis, AWS

- Reduced average API response time from over 2 minutes to under 1 second by optimizing PostgreSQL queries, reducing expensive joins, and implementing pagination, indexing, and Redis caching.
- Improved backend scalability, security, and reliability while upgrading the application from Node.js 20 to Node.js 24 LTS.
- Optimized API architecture and data access patterns to reduce unnecessary database operations and improve overall application performance.

Enterprise Scientific Publishing Platform — US

Next.js, Sanity.io, Tailwind CSS, Shadcn, Vercel

- Led migration of an enterprise-scale scientific publishing platform to Sanity.io, covering content modeling, migration strategy, database considerations, and feature migration.
- Designed a scalable headless CMS architecture using Next.js and Sanity.io while preserving existing journals, content relationships, and data integrity.
- Migrated and rebuilt existing functionality while maintaining a consistent publishing workflow and modernizing content delivery.

AI Voice Agents & Automation Platform

Next.js, n8n, ElevenLabs, Vercel, Google APIs

- Developed AI-powered voice agents for healthcare providers to automate appointment booking, confirmations, reminders, rescheduling, and calendar management.
- Built end-to-end automation workflows integrating ElevenLabs, n8n, Google Calendar, and Google Sheets.

- Developed a back-office portal for managing conversations, call transcripts, AI-generated summaries, scheduling data, and analytics.

Hotel Management System

NestJS, Node.js, PostgreSQL, Redis, React.js, React Native, AWS

- Led development of a complete hotel management platform and delivered web and mobile applications within 6 weeks.
- Designed scalable backend services and REST APIs using NestJS, PostgreSQL, and Redis for booking, inventory, and role-based management.
- Deployed application infrastructure on AWS and collaborated with frontend and mobile teams to deliver the complete solution.

Coaching Platform — Saudi Arabia

Django REST Framework, PostgreSQL, django-tenants, Celery, Next.js, React Native, Expo, AWS CDK, ECS Fargate

- Developed core platform capabilities including instant calling and scheduled calls for a coaching marketplace.
- Contributed across backend, web, mobile, and cloud infrastructure, working with a multi-tenant architecture and asynchronous background processing.
- Built and integrated REST APIs supporting the web and mobile applications.

ACHIEVEMENTS

- AI Hackathon 2025 Winner — Team Lead
- Reduced API response time from over 2 minutes to under 1 second
- Improved page-load performance from 15 seconds to approximately 2 seconds
- Delivered a complete hotel management platform across web and mobile in 6 weeks
- Successfully migrated an enterprise scientific publishing platform to a headless CMS architecture

EDUCATION

BS, Computer Science **August 2016 – August 2020**

Mohammad Ali Jinnah University (MAJU), Karachi — GPA: 3.2

CERTIFICATIONS

- Namaste Frontend System Design (In Progress) — Namaste Dev
- Namaste Node.js (2025) — Namaste Dev
- Namaste React: From Zero to Hero (2024) — Namaste Dev
- Clean Code (2023) — Udemy
- React Native: The Practical Guide (2023) — Udemy